

Modeling the Spread of COVID-19 Using a Logistic Growth Model in MATLAB

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Introduction: Logistics Growth Model & Uploading Data

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Logistic Growth Model Basics

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The logistic model used is defined as:

$$N(t) = \frac{K}{1 + e^{-r(t-t_0)}}$$

Where:

- $N(t)$: cumulative number of cases at time t
- K : carrying capacity (*Indicates the estimated maximum number of cases the country may reach under current conditions*)
- r : growth rate (*Reflects how quickly the cases are increasing*)
- t_0 : inflection point (*peak growth time, the time when the curve switches from accelerating to decelerating growth*)

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The Numerical Engine: Implementing the Levenberg-Marquardt Algorithm (Using `lsqcurvefit` function)

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Levenberg-Marquardt Algorithm

$$\mathbf{p}_{k+1} = \mathbf{p}_k + (\mathbf{J}^T \mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{J}^T \mathbf{r}(\mathbf{p}_k)$$

$\mathbf{p}_{k+1}$

This is the **new, improved guess** for the parameters after taking one step.

$\mathbf{p}_k$

This is the **current guess** for the parameters before taking one step.

$\mathbf{J}$

This matrix tells us **how sensitive the model's output is** to small changes in each parameter.

$\mathbf{r}(\mathbf{p}_k)$

This is a list of the **errors** between the actual data and the model's prediction using our current guess.

λ

This is the **"smart" control knob** that adjusts the size and safety of the step the algorithm takes.

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Step – 01: The Starting Point

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⇒ **Purpose:** Provide an Initial Guess (p0), based on the data

⇒ $K(\text{Carrying Capacity}) = \max(N_{\text{fit}}) * 1.5;$

⇒ $r(\text{rate}) = 0.1;$

⇒ $t_0(\text{Inflection}) = t_{\text{fit}}(\text{round}(\text{end}/2));$

```
659 - function FitModelButtonPushed(app, event)
660 -     if isempty(app.Country1Data)
661 -         app.StatusLabel.Text = 'X Please select Country 1 first';
662 -         app.StatusLabel.FontColor = app.errorColor;
663 -         return;
664 -     end
```

```
685 - 
686 -     % Initial parameter estimates
687 -     K_init = max(N_fit) * 1.5;
688 -     r_init = 0.1;
689 -     t0_init = t_fit(round(end/2));
690 -     initial_params = [K_init, r_init, t0_init];
691 -
```

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Step – 02: Constraining the Solution Space

⇒ Purpose:

- Ensure physically meaningful results
- Improve efficiency

⇒ Bounds:

Lower:

- K must be at least the max observed cases.
- r must be positive.

Higher:

- Provides a reasonable ceiling to prevent runaway solutions..

```
691  
692  
693 - % Parameter bounds  
694 - lower_bounds = [max(N_fit), 0.01, 0];  
695 - upper_bounds = [max(N_fit)*10, 1, max(t_fit)*1.5];
```

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Step – 03: Executing The Solver

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Purpose:

- To execute the **Levenberg-Marquardt algorithm** with our specified instructions.

```
% Curve fitting
options = optimoptions('lsqcurvefit', 'Display', 'off', 'MaxIterations', 1000);
[fitted_params, ~, ~, exitflag] = lsqcurvefit(...
    logistic_model, initial_params, t_fit, N_fit, lower_bounds, upper_bounds, options);
```



The `lsqcurvefit` function takes

- The model function handle (`logistic_model`)
- The initial guess (`initial_params`)
- The data (`t_fit`, `N_fit`)
- The constraints (`lower_bounds`, `upper_bounds`)

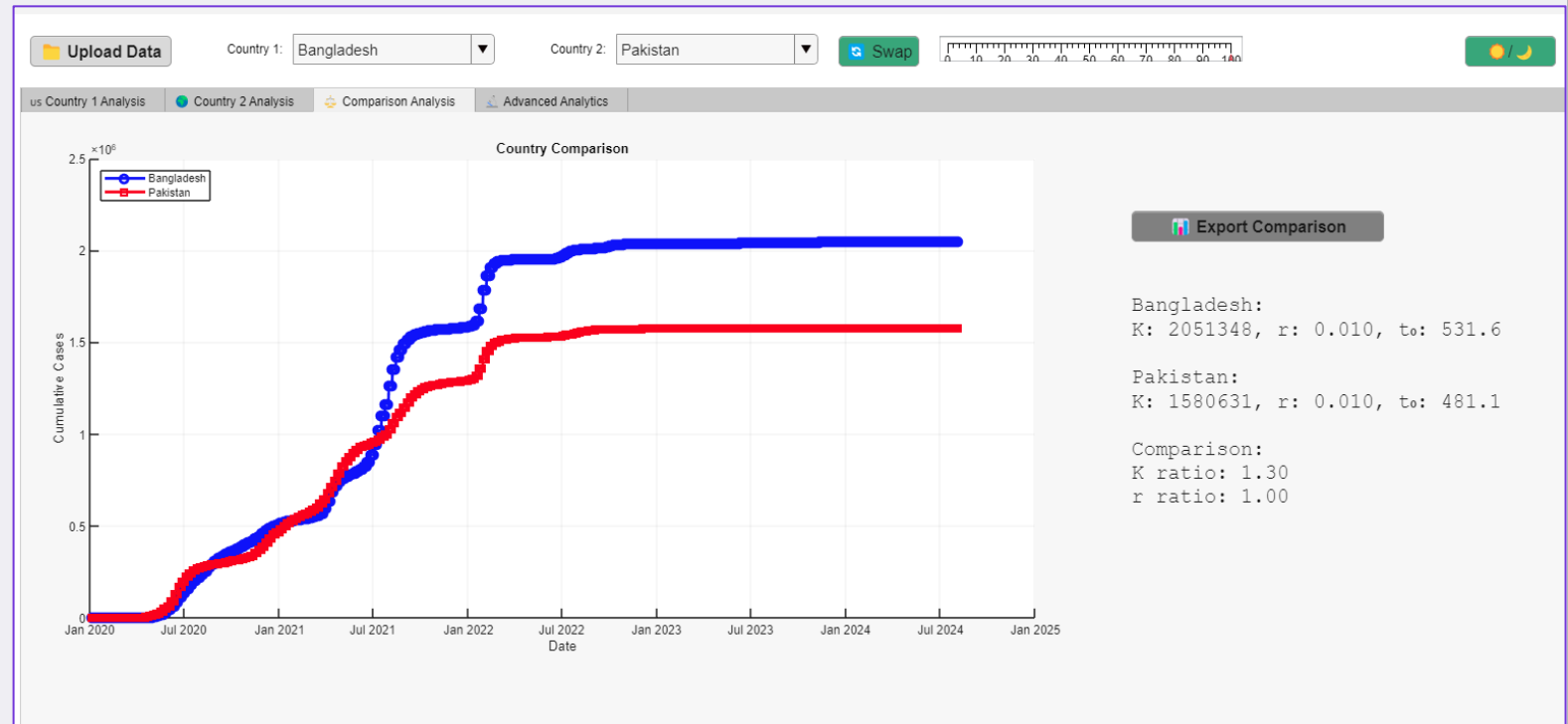
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The Comparison Tab

Functionality:

- By solving for two countries, we move from simple calculation to analytical insight.
- The "Comparison" dashboard visualizes both solutions simultaneously.
- Enables direct comparison of key parameters (K , r , t_0) and pandemic trajectories.



	Visualizing and Validating The Numerical Solution	
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- How good the curve fit is?
- **Our Validation Tool:** The app's visual dashboard is generated by the *plotModelResults* function.
- **Two-Step Validation Process:**
 - Direct visual comparison of the model curve against the actual data.
 - Numerical validation by analyzing the errors, or "residuals."

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What We Plot:

Blue Circles: The actual, raw data points from the CSV file.

Pink Line: Our model's curve fitted to the data.

The Goal: To see how closely the red line tracks the blue circles.

The Result: A close alignment between the line and the data gives us our first confirmation that the model is a good fit.

Code Location: [plotModelResults](#) function.

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Residual:

$$\text{Residual} = (\text{Actual} - \text{Predicted})$$

- **Why Plot Them?:** To find hidden patterns in the model's errors.
- **What We Want to See:** A random scatter of points centered around the zero line.
- **What We Want to Avoid:** Any clear pattern (like a U-shape or a trend), which would indicate a flaw in our model.

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The PDF Report Exporter:

A one-click "Generate PDF Report" button in the GUI.

Function: `GeneratePDFReportButtonPushed`

The Key command: `exportgraphics`

It captures all the plots from our analysis.

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Advanced Analytics - GUI
explanation from

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⇒ Country dashboard:

Here user can see a selected country's Actual vs Logistic model, Residuals, Growth rate and K parameter Sensitivity

The key feature is

Growth rate - The growth rate r is a parameter in the logistic model that describes how fast the number of cumulative COVID-19 cases increases over time during the outbreak.

K parameter Sensitivity - It means how changing the value of K affects the shape and behaviour of the logistic curve, while keeping the other parameters fixed

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⇒ World dashboard:

Same as country dashboard where it shows data for the whole world.

In the show matrices –

we can see the max number of infected persons by carrying capacity.

RMSE - measures the average size of the errors between the model's predictions and the actual COVID-19 case data

$$RMSE^2 = \frac{1}{n} \sum_{i=1}^n (y_i - y_{i(pred)})^2$$

R^2 = how much of the real-world data the model captures.
Greater R^2 means better fitted.

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⇒ Date statistics:

It takes date as input and shows *Total Infected* and *Total Deaths*

The code

```
function showDateStats(app)
```

```
    dtxt = app.AdvDateField.Value;
```

[Reads the text input from the date field]

```
    try
```

```
        target = datetime(dtxt, 'InputFormat','yyyy-MM-dd'); [Converts the input string into a datetime object]
```

```
    Catch [ If the conversion fails catch block runs. ]
```

```
    if ~isempty(app.AdvResultLabel)
```

```
        app.AdvResultLabel.Text = 'Use YYYY-MM-DD format';
```

```
        app.AdvResultLabel.FontSize = 20;
```

```
        if app.isDarkMode
```

```
            app.AdvResultLabel.FontColor = [1 1 1];
```

```
        else
```

```
            app.AdvResultLabel.FontColor = [0.1 0.1 0.1];
```

```
        end
```

```
    end
```

```
    return;
```

```
end
```

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⇒ Compare countries:

Can compare two countries and observes the

logistic parameters
most fitted curve
saturation

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→ Saturation Date:

Finds saturation date for the first country

Code:

```
data=extractCountryData(app,c);           takes country-specific COVID-19 data
pars=fitLogistic(app,data);   fits the logistic model
K=pars.K;   Defines the logistic function inline
logistic=@(p,t) p(1)./(1+exp(-p(2)*(t-p(3))));
t_future = linspace(0, max(data.TimeDays)*2, 2000);   Extends time into the future to simulate the full curve (2 × current duration)
curve = logistic([pars.K pars.r pars.t0], t_future);   Computes the logistic curve values over time.
sat95 = 0.95*pars.K;   Calculates the 95% saturation threshold.
idx = find(curve>=sat95,1);   Finds the first time index when curve reaches 95% of K.
date_sat = data.Date(1)+days(t_future(idx));   converts the predicted time index back into a calendar date
if ~isempty(app.AdvResultLabel)
app.AdvResultLabel.Text = sprintf('%s\nSaturation (95%%)=%.0f\nPredicted Date: %s',c,sat95,datestr(date_sat));
app.AdvResultLabel.FontSize = 20;
if app.isDarkMode
app.AdvResultLabel.FontColor = [1 1 1];
else
app.AdvResultLabel.FontColor = [0.1 0.1 0.1];
end
end
end
```

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⇒ Critics and understand the limitations:

While the logistic growth model provides a simple and interpretable way to analyze the cumulative spread of COVID-19, it comes with important limitations.

- Single-Wave Assumption
- Fixed Parameters
- No Real-Time Adaptability
- Ignores External Factors
 - Vaccination campaigns
 - Virus variants (e.g., Delta, Omicron)
 - Travel restrictions or reopening policies
- Oversimplification
- Assumes Accurate Data
- Fails in Early/Flat Regions

As a result, its predictions may lose accuracy in highly dynamic or unpredictable settings. Residual patterns and parameter sensitivity help reveal these limitations, highlighting the need for caution when interpreting long-term forecasts.

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⇒ Tentative way to overcome some limitations

- Using Multi-Wave Models
- Incorporate External Factors
- Switch to Epidemiological Models
- Hybrid Modelling

Incorporating these models and factors will lead to more accurate and realistic predictions.

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The end
Thank you

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For any queries

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